

Day 2 - Even/Odd Number Check

Question

Ek number diya gaya hai.

Check karo ki number **Even** hai ya **Odd**.

Example

Input

8

Output

Even

Input

7

Output

Odd

Approach 1 - Using Modulus Operator (%)

Logic

Agar number ko 2 se divide karne par remainder 0 aata hai to Even.

Warna Odd.

Dry Run

N = 8

$8 \% 2 = 0$
Even

N = 7

$7 \% 2 = 1$
Odd

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Java Code

```
public class Day2 {  
    public static void main(String[] args) {  
  
        int n = 8;  
  
        if(n % 2 == 0)  
            System.out.println("Even");  
        else  
            System.out.println("Odd");  
    }  
}
```

Approach 2 - Using Bitwise AND (&)

Logic

Binary me:

Even Number ka last bit = 0

Odd Number ka last bit = 1

Agar

$n \& 1 == 0$

to Even.

Dry Run

N = 8

```
1000
&
0001
----
0000
```

Result = 0

Even

N = 7

```
0111
&
0001
----
0001
```

Result = 1

Odd

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Java Code

```
public class Day2 {  
    public static void main(String[] args) {  
  
        int n = 7;  
  
        if((n & 1) == 0)  
            System.out.println("Even");  
        else  
            System.out.println("Odd");  
    }  
}
```

Approach 3 - Using Ternary Operator

Logic

Short form of if-else.

Java Code

```
public class Day2 {  
    public static void main(String[] args) {  
  
        int n = 10;  
  
        String result = (n % 2 == 0) ? "Even" : "Odd";  
  
        System.out.println(result);  
    }  
}
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Interview Follow-Up Questions

Q1. Negative numbers ke liye bhi kaam karega?

Yes.

$-4 \% 2 = 0$

Even

$-5 \% 2 = -1$

Odd

Q2. Fastest Method?

Bitwise AND (&)

$(n \& 1) == 0$

Kyunki divide/modulus operation nahi hota.

Q3. Interview me kaunsa answer dena chahiye?

Pehle:

$n \% 2 == 0$

Fir interviewer ko batao:

$(n \& 1) == 0$

Bitwise approach bhi hoti hai.

Ye impress karta hai.

Summary

Approach	Time	Space
Modulus (%)	O(1)	O(1)
Bitwise (&)	O(1)	O(1)
Ternary	O(1)	O(1)

Aaj Ka Interview Concept

Modulus vs Bitwise

`10 % 2 == 0`

Easy aur readable.

`(10 & 1) == 0`

Low-level aur thoda faster.

Practice

Check karo:

15
22
99
100
101

Output khud predict karo.